

Optical Spectroscopy of Free-Propagating Plasma and Its Interaction with Tungsten Targets in PF-1000 Facility

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The paper describes the application of optical spectroscopy methods for investigation of intense plasma streams, which are emitted from the large PF-1000 facility operated at 21–27 kV, 290–480 kJ, as well as for studies of interactions of such streams with tungsten (W) targets. The electron concentration, as estimated from the Stark broadening of the D_β line recorded at a distance of $z = 30$ cm from the electrode ends, changed from about 2.2×10^{16} to 0.5×10^{16} cm⁻³ in about 5 μ s. Spectroscopic measurements were also carried out in experiments performed with a pure W-target placed at $z = 30$ cm. There were for the first time in PF-1000 recorded and identified many WI and WII lines. In some cases an analysis of the D_β line was impossible because of its very strong re-absorption. A quantitative analysis of the W-lines was not performed, but some information about dynamics of the W-ions production was obtained.

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1 Introduction

Research on high-current discharges of the Plasma-Focus (PF) type has been carried out in different laboratories for many years [1–4]. Detailed studies have concerned dynamics of PF discharges and the emission of X-rays, ions, electrons, and D-D fusion products (fast neutrons and protons), but optical spectroscopy has been used rarely. Some preliminary spectroscopic measurements were performed within the large PF-1000 facility [4], taking into account that information about intense plasma streams during their propagation or interaction with different targets is of interest for science and technology. Experiments with composed C-W targets within PF-1000 facility showed that intense deuterium lines and C-lines can mask the tungsten emission, and the tungsten lines could hardly be identified [5]. Therefore, it was decided to perform experiments with pure W-targets using intense laser beams [6] or plasma streams from other facilities [7]. In the mean-time a current pulse generator of the PF-1000 facility was modernized and its operational parameters were improved, and in particular oscillations of the discharge current were reduced, reproducibility of discharges and neutron yields were increased. Hence, the main aim of recent studies was to perform new spectroscopic measurements of free-propagating plasma streams and their interactions with pure W-targets.

2 Experimental setup

During the described spectroscopic studies the PF-1000 machine [4] was equipped with coaxial electrodes of 560 mm in length. The inner one was a 230-mm-dia. copper tube, and the outer one consisted of twelve 80-mm-dia. SS-tubes distributed upon a 400-mm-dia. cylinder. After pumping out the chamber to the basic pressure of about 10^{-5} hPa, and filling it with a pure D₂-gas under pressure $p_0 = 0.75$ –2.1 hPa, the triggered discharges were supplied from a 1.32-mF condenser bank charged to 21–27 kV, 290–480 kJ. The maximum discharge current depended on the initial charging voltage and it amounted to 1.5–1.8 MA.

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The recent spectroscopic measurements of the free plasma streams have been carried out perpendicular to the z -axis and at a distance $z = 30$ cm from the electrode outlet. Spectroscopic measurements were performed behind a quartz window, through an optical collimator coupled with an optical fiber-cable and a Mechelle®900 spectrometer, which enabled the exposition time from 100 ns to 100 ms to be chosen. To investigate plasma produced from W-targets the use was made of a movable support, which could be inserted vertically at $z = 30$ cm. A general view of the experimental arrangement is shown in Fig. 1.

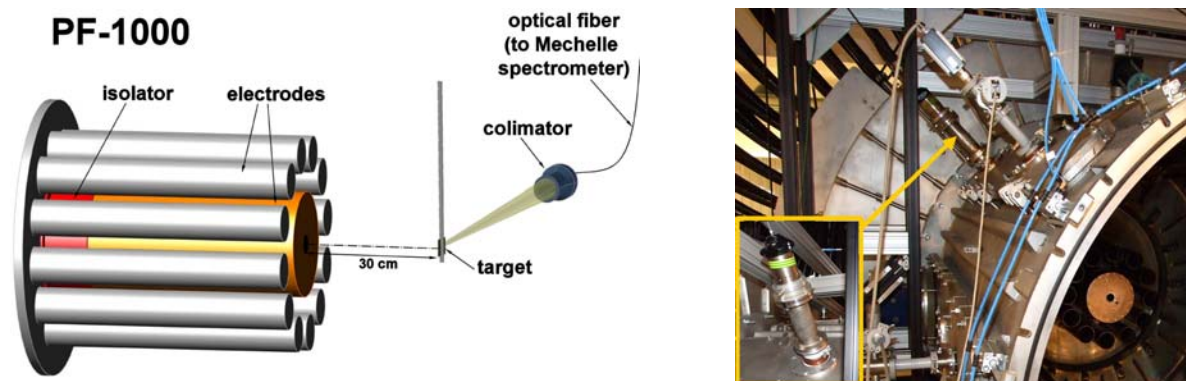


Fig. 1 Scheme of the experimental arrangement and positioning of the quartz window and collimator for spectroscopic measurements within the PF-1000 facility. (Color figure: www.cpp-journal.org).

3 Experimental results

The described studies were initiated by taking time-integrated pictures of investigated plasma streams by means of red filters and CCD camera placed behind the observation window. Some exemplary pictures are presented in Fig. 2.

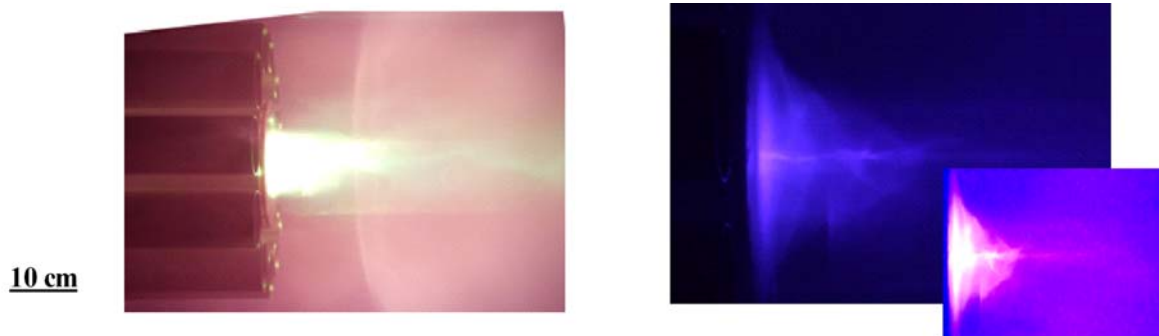


Fig. 2 Time-integrated pictures of plasma streams in PF-1000, as taken for 2 different shots: left - shot #8356 (no filter), $p_0 = 0.75$ hPa D_2 , $Y_n = 9 \times 10^{10}$; right - shot #8368 (red filter), $p_0 = 0.7$ hPa $D_2 + 0.05$ hPa Ar, $Y_n = 5.1 \times 10^9$. (Color figure: www.cpp-journal.org).

The time-integrated pictures have shown that spatial structures and dimensions of plasma streams depended considerably on the working gas and its pressure. For further studies the use was made of the pure deuterium filling only. The emission spectra were recorded with the Mechelle®900 spectrometer at different instants of the discharge, as shown in Fig. 3.

The recorded spectral lines were analyzed with the use of a Kurucz database [8]. It was confirmed that at the pure D_2 -filling the PF-1000 facility emits first the deuterium-plasma stream and one can observe intense deuterium Balmer lines, but at a distance $z = 30$ cm about $2.5 \mu s$ there appear many impurity lines originating mainly from the Cu-electrodes, i.e. Cu- lines. In the analysis particular attention was paid to the D_β line, as shown in Fig. 4.

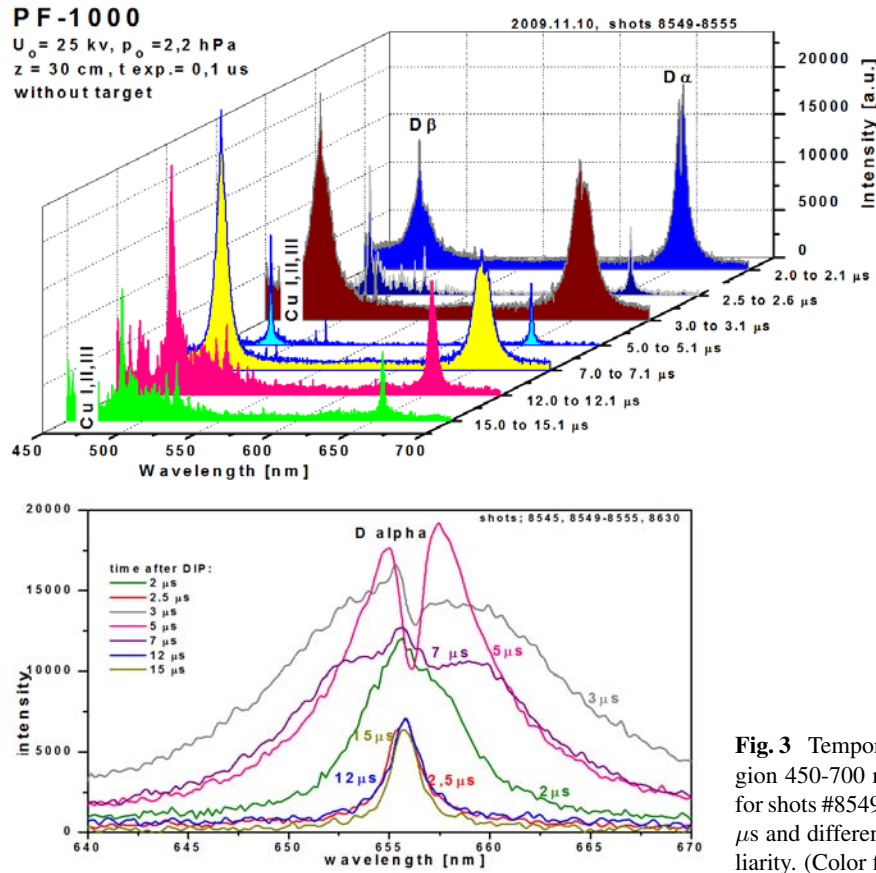


Fig. 3 Temporal changes of spectrum in the region 450-700 nm and D_α profiles, as recorded for shots #8549-8555, with the exposition of 0.1 μs and different instants after the current peculiarity. (Color figure: www.cpp-journal.org).

It should be noted that, in spite of the careful calibration, the observed ratio of D_α and D_β intensities was different from the value characteristic for plasma in a thermal equilibrium, what could be explained by a lack of the full thermalization. Nevertheless, on the basis of the recorded D_β line it was possible to estimate the electron density value. Considering the Stark broadening and using the known formula $n_e = A \Delta\lambda^B$, where according to H.R. Griem [9] $A = 1.02 \times 10^{16}$ and $B = 1.45$ or according to C.R. Vidal [10] $A = 1.17 \times 10^{16}$ and $B = 1.46$, $\Delta\lambda^B$ - the measured Stark broadening, it was found that at the observation plane the electron concentration changed from about 1.8×10^{16} to $0.8 \times 10^{16} \text{ cm}^{-3}$ in about 14 μs , as shown in Fig. 4 (right).

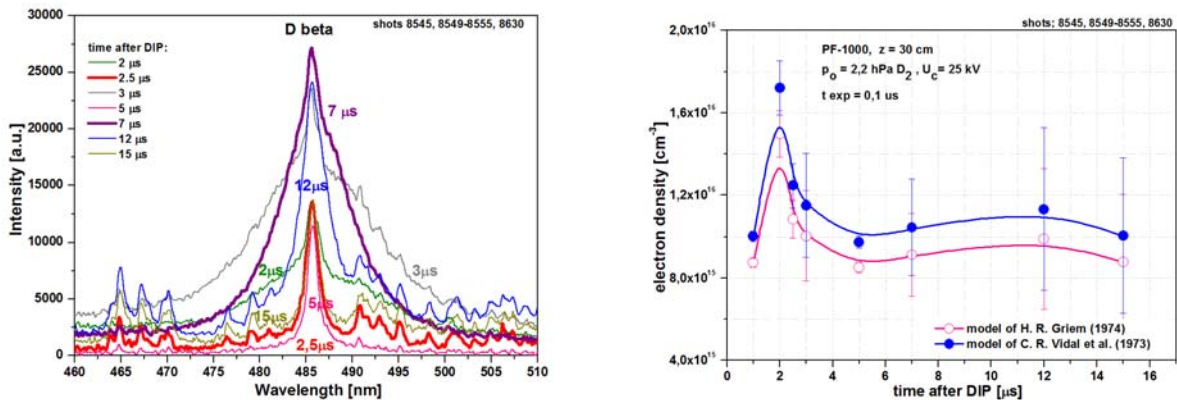


Fig. 4 D_β profiles recorded for different instants after the current dip and estimated values of the electron density in the free-propagating plasma stream at $z = 30 \text{ cm}$. (Color figure: www.cpp-journal.org).

The second part of the PF-1000 experiment concerned spectroscopic measurements of a "plasma plum" produced by the D-plasma stream irradiating a target made of the pure tungsten, which was placed at $z = 30$ cm. Some exemplary spectra are presented in Fig. 5.

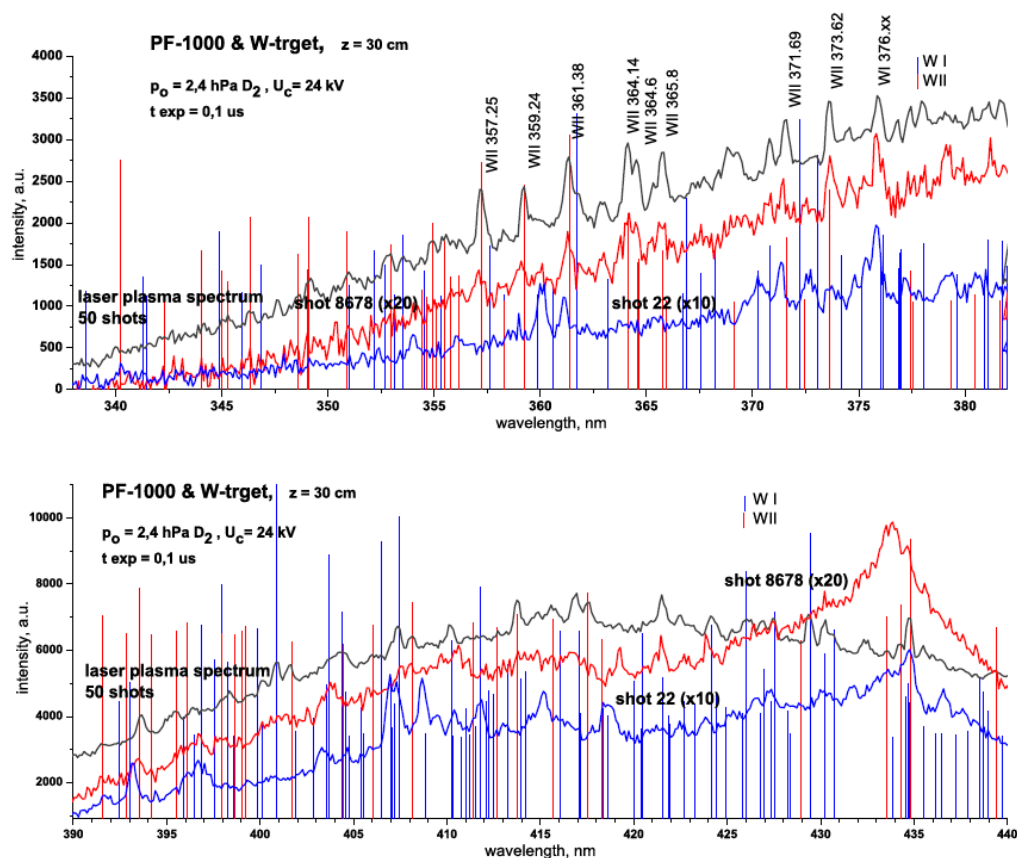


Fig. 5 Comparison of spectra obtained from W-targets in the laser experiment (top trace) and two shots within PF-1000 (middle and bottom traces). To facilitate the identification of WI and WII lines the spectra obtained from those shots have been presented with different amplification, and data taken from the Kurucz database [8] were shown by thin vertical lines. (Color figure: www.cpp-journal.org).

The presented spectra show that in the PF-1000 experiment the plasma plum produced from the W-target have emitted different WI and WII spectral lines, which were observed in the earlier laser-target experiment [6, 11]. Unfortunately, the detailed analysis could hardly be performed because of rich tungsten spectra and the intense continuum emission. Comparing the spectra presented in Fig. 5, which were obtained with the same exposition time equal to $0.1 \mu\text{s}$, but at different instants (bottom trace - $6.5 \mu\text{s}$ after dip, middle trace - $7.1 \mu\text{s}$ after dip), one can notice that relative intensities of some WI lines observed in the earlier phase decreased in the considered period (from 6.5 to $7.1 \mu\text{s}$). In contrary, relative intensities of some distinct WII lines (e.g., WII-364.14, 364.6 and 365.8) increased during that period considerably. It informed about dynamics of the W-plasma production and expansion. In fact, that effect was observed until the maximum intensity of the WII lines was recorded. Information about W-ions with a higher ionization could not be collected because of limitations of the applied optical spectrometer and a lack of data (about WIII and higher W-lines) in the Kurucz and other databases. An analysis of the D_β line was impossible because of the very strong re-absorption effect. Taking into consideration that the local thermal equilibrium within the free-propagating plasma stream as well as the thermalization process during the irradiation of the W-target was questionable, it was impossible to estimate electron temperatures of D- and W-plasma during the whole experiment.

The erosion of the W-target was estimated by an inspection of the irradiated sample, which was performed after its removal from the PF-1000 chamber. The use was made of an optical microscope and SEM method. It

has shown the considerable erosion of the whole target surface. Using an electronic balance, it was determined that a W-target (of the total surface equal to 32 cm², weighting 10 g, and placed at a distance of 30 cm) lost on average about 1 mg during a single PF-1000 shot. That loss was caused by the plasma stream of the power flux equal to about 10⁹ W/cm². In order to determine the ion streams, which irradiated the investigated W-targets in the PF-1000 facility, there were performed additional measurements of ion beams by means of pinhole cameras and nuclear track detectors, which are reported in another paper [12].

4 Summary and conclusions

The most important results of this study can be summarized as follows: 1. The use of optical spectroscopy made it possible to investigate the emission of spectral lines from the D-plasma stream and from the W-target; 2. The values of the electron densities, as estimated on the basis of the D _{β} line analysis, remain in agreement with results obtained by other techniques, e.g. laser interferometry [12]; 3. A quantitative analysis of the recorded W-lines could hardly be performed, but some information about dynamics of the W-ions production was obtained; 4. The erosion of the W-target by the intense D-plasma streams was estimated from the surface analysis and the target weighting. One can conclude that the spectroscopic studies and particularly time-resolved measurements should be continued in order to collect more information about plasma dynamics and its interactions with solid targets.

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